

Self-test Exercise

1. A _____ variable is used to incorporate qualitative information in a regression model.
 - a. dependent
 - b. continuous
 - c. binomial
 - d. dummy

2. Which of the following models is appropriate when the dependent variable is binary?
 - a. Probit model
 - b. Logit model
 - c. LPM
 - d. All of the above

3. Which of the following regression models that its coefficients are estimated using maximum likelihood estimation?
 - a. Probit model
 - b. Logit model
 - c. LPM
 - d. (a) and (b)

4. Which of the following is an example of a binary response model?
 - a. Moving average model
 - b. Autoregressive model
 - c. FGLS model
 - d. Logit model

5. In the probit model, G is the:
 - a. standard normal cumulative distribution function.
 - b. cumulative distribution of normal distribution function.
 - c. standard normal probability distribution function.
 - d. normal probability distribution function.

6. Which of the following statements is true regarding the marginal effects calculated in the probit model and the logit model?
 - a. Partial effects in the probit model are easier to interpret than in the logit model.
 - b. Partial effects in the logit model are easier to interpret than in the probit model.
 - c. Partial effects in both models are difficult to interpret because they depend on the value of the independent variable.

d. Both models do not allow for calculation of partial effects.

7. The difference between the LPM model and the logit and probit models is that:

a. the LPM assumes constant marginal effects for all the independent variables, while the logit and probit models imply diminishing magnitudes of the partial effects.

b. the LPM assumes constant marginal effects for some of the independent variables, while the logit and probit models imply diminishing magnitudes of the partial effects.

c. the LPM assumes constant marginal effects for the dependent variable, while the logit and probit models imply diminishing magnitudes of the partial effects.

d. the LPM assumes different marginal effects for all independent variables, while the logit and probit models imply diminishing magnitudes of the partial effects.

8. The parameters in a linear probability model can be interpreted as measuring the change in the probability that $y = 1$ due to a one-unit increase in an explanatory variable.

a. True

b. False

9. The cumulative distribution function for a standard logistic random variable is a decreasing function.

a. True

b. False